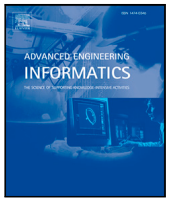
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Full length article



A digital twin-based motion forecasting framework for preemptive risk monitoring☆

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ABSTRACT

Risk monitoring is a critical task in numerous industrial fields, including construction engineering. Existing approaches primarily focus on identifying the immediate incidents of unsafe events, often neglecting the latent risks arising from the high uncertainty of agents' motions. To bridge this gap and lay the basis for preemptive risk monitoring, this paper proposes a novel and effective digital twin-based motion forecasting framework that leverages cloud-edge collaboration to predict the future trajectories of workers on construction sites, while ensuring data privacy and reducing computational burden. The edge-end acquires multi-view images about the physical scene and extracts multi-scale features with a deep learning-based backbone. In the cloud-end, where the digital world is hosted, a Transformer-based trajectory tracking model is developed to construct a behavioral digital replica comprising of multiple workers' historical trajectories from the bird's-eye view perspective. Subsequently, a LSTM-based model, combined with the social pooling mechanism, is deployed to obtain spatiotemporal interactive features from the historical trajectories and predict workers' future motions. Experiments are conducted on a dataset (WHUT-MF) collected from real-world constructions sites. Our framework achieves 0.528 m in ADE and 0.84 m in FDE on entire WHUT-MF, which demonstrate that the proposed digital twin-based motion forecasting model accurately predicts the future trajectories of workers in various scenarios with potential risks.

1. Introduction

Risk monitoring is a crucial and challenging issue in complex and highly dynamic scenes such as construction sites. These sites consist of numerous heterogeneous entities, also referred to as agents, such as building materials, heavy machinery and workers, which interconnect with each other and affect the security of construction sites collectively. Existing risk monitoring approaches typically focus on identifying the current safety state of construction sites through the recognition of workers' behaviors and machinery operational states [1, 2]. However, in complex scenes involving heterogeneous and dynamic agents, evaluating the current state alone is insufficient. It is essential to assess the latent risks arising from motion uncertainties and the complex interaction between diverse agents. Such assessments allow for proactive risk monitoring and timely intervention to prevent disastrous safety consequences, such as entering restricted areas or colliding with moving obstacles. Workers are the primary category of dynamic agents and the focal point of safety management on construction sites. Their motions are highly dynamic, driven both by task-oriented intention and context-related interaction. Therefore, accurately predicting their motion trends, i.e. future trajectories, is challenging but vital for preemptive risk monitoring on construction sites.

The motion of workers is driven by their individual intentions towards certain targets, as well as their interactions with the surrounding

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environment. Most mechanism-based human motion forecasting approaches [3,4] usually employ predictive optimal control or plan-based methods to inference a probabilistic model of human motion trajectories based on historical movements. This means they predominantly consider motion information but lack a detailed understanding of the surrounding contexts. They also struggle to handle the unpredictable and complex interactions between agents. On construction sites, agents' movements are influenced by a multitude of diverse environmental factors, making it challenging for mechanism-based approaches to forecast human motion accurately. To improve the accuracy and robustness of motion forecasting in dynamic scenes, data-driven learning methods have been extensively explored in domains such as computer vision [5,6], intelligent transportation [7,8], etc. In comparison, there is still limited research in the context of construction sites. To predict the future trajectories of workers using data-driven methods in construction site scenarios, acquiring a massive amount of historical motion data is necessary. Furthermore, it is necessary to establish refined representations of large-scale scenes, enabling the encoding of complex scene dynamics and interactive relations to realize accurate motion prediction.

However, the aforementioned work faces significant challenges due to the following reasons: (1) The most commonly adopted sensors on construction sites are cameras typically deployed independent of one another, making it difficult to obtain a comprehensive fine-grained representation of large-scale scenes. (2) The limited computational power of onsite computing platforms hinders real-time execution of deep learning-based motion prediction model. Moreover, the platform's ability to support the upstream tasks of motion prediction, namely object detection and tracking, is even more demanding. The prevalent approach adopted in recent years has been to transmit the collected data from construction sites to the cloud for processing. Nevertheless, this practice is unfavorable for privacy protection and data security, while also imposing significant pressure on data transmission.

To address the aforementioned issues, our work proposes leveraging the digital twin technology, which has been widely utilized to simulate and model physical counterparts, capturing their physical details and visual features. For example, numerous researches [9–11] adopt digital Twin for the purpose of building information modeling (BIM) to optimize the management of the production process. However, for certain tasks, such as motion prediction, what is required is not limited to observable information of the physical counterpart, but also implicit and intricate features.

In this paper, we propose an edge-cloud collaboration framework to implement digital twin-based motion forecasting for risk monitoring on construction sites. At the edge-end, a pre-trained ResNet is utilized as the backbone network to extract multi-view feature maps at each of the three edge-computing platforms we deployed in the construction sites, for the purpose of privacy protection, data security and lower transmission burden. The cloud-end hosts the digital twin in which a multi-object tracking and a motion forecasting model is deployed. Specifically, inspired by previous works [12–15], a Transformer-based multi-object tracker is devised to exploit the cross-attention mechanism, so as to capture the expressive motion characteristics of multiple agents across multi-view features transmitted from the edge-end. Historical trajectories are then obtained by decoding the learned query and utilized to form a fine-grained task-oriented behavioral digital replication of the scene. Subsequently, a social LSTM-based [16] motion forecasting model is deployed to obtain spatiotemporal interactive features among the multi-agents, which are then decoded to produce the predicted future trajectories of agents.

The main contributions of our work can be summarized as follows.

1. This study proposes a cloud–edge collaborative framework for constructing a digital twin aiming to perform motion forecasting of workers. At the edge-end, multi-scale features are extracted from the acquired multi-view images for privacy protection and data security. The cloud-end hosts the digital realm in which a Transformer-based multi-object tracking network and a social LSTM-based motion forecasting model are deployed to obtain historical trajectories and furthermore predict future trajectories of multiple workers.

2. To create a task-centric digital replication of the physical scene, a query-based tracking model is developed to obtain historical trajectories of multiple objects across multiple views. Building upon this, we employ social LSTM to capture the spatiotemporal interaction features among the agents, ultimately predicting high-precision future trajectories for each and every worker.

3. The experimental results, conducted on a dataset collected from real-world construction sites, demonstrate that the proposed digital twin-based motion forecasting model can accurately predict the future trajectories of workers in various scenarios with latent risks.

2. Related work

2.1. Risk monitoring on construction sites

Prompted by the rapid development of Artificial Intelligence (AI) algorithms and their successes in computer vision, vision-based approaches for risk monitoring have become a research focus in construction engineering [17–19]. These approaches typically focus on detecting the risks stemming from unsafe behaviors of workers [20–22] such as not wearing helmets [23–25], falling from heights [26–28], and collisions with safety-related objects [29,30], etc. However, evaluating the immediate safety status of the scene through detection-based methods is inadequate for handling fast-evolving and stochastic potential risks, promoting the development of proactive prediction-based approaches. Several studies have approached risk prediction through motion forecasting, primarily utilizing model-based or mechanism-based methods: Rashid [31] developed a physics-based human trajectory prediction model for evaluating collision risks around hazard areas, Amir [32] studies the feasibility of the proactive safety framework by utilizing two mechanism-based trajectory prediction models via field experiments. A significant limitation lies in the difficulty of rule-based approaches to adapt to complex environments. For scene-adaptive and robust motion forecasting, we propose a deep learning-based framework, enabled by the digital twin technology, in the service of preemptive risk monitoring.

2.2. Motion forecasting methods

Motion forecasting serves as the foundation for preemptive risk monitoring by predicting the motion trends of dynamic agents, which are inherently influenced by both the environmental conditions and individual behavioral intentions. Several motion forecasting studies [33–36] employ recurrent neural networks (RNNs) to iteratively capture the temporal evolution patterns of agents' motion for prediction. To model the complex and diverse interactive relationships among agents, other research works [16,37,38] employ convolutional neural networks (CNNs) to capture spatial interactions based on the encoded historical motions with RNNs. Recent studies [39,40] propose to integrate the spatiotemporal encoding of dynamic and static elements in the environment, utilizing cross-attention mechanisms to learn intricate interactions between the agents and scene context. Furthermore, generative models such as Generative Adversarial Networks (GANs) [41] and Gaussian Mixture Models (GMMs) [42] have also been introduced for motion forecasting.

Some studies have focused on motion forecasting of workers on construction sites, where explicit geometric and topological structures may be absent. Therefore, they emphasize the use of historical motion and the modeling of interactions between dynamic agents via Generative Adversarial Networks [43] and recurrent neural network based methods [44,45].

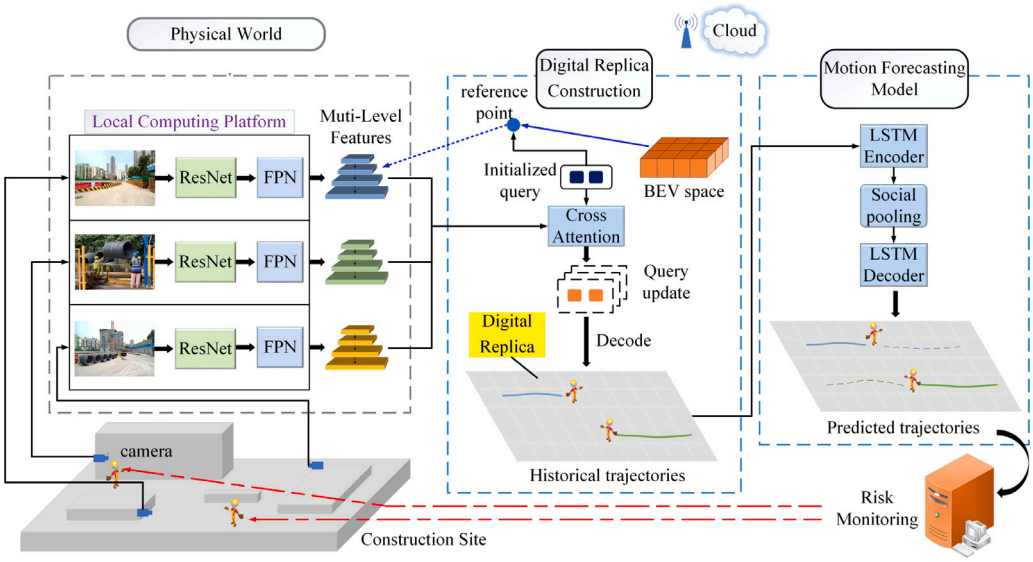


Fig. 1. Overall architecture of the proposed digital twin based motion forecasting framework.

2.3. Digital twin-based applications in construction engineering

The essence of a digital twin is to establish a task-oriented replication of the physical world as the digital realm [46–49]. As a universal technical framework, it has been incorporated into the whole life cycle of various infrastructural entities [50–52] such as machinery and buildings. Utilizing the digital twin technology, the construction community proposes numerous approaches to construct fine-grained 3D digital replications of infrastructures to implement effective and efficient construction progress management, thereby empowering digital fabrication [53–58]. In these studies, the construction of the digital twins focus primarily on replicating explicit and visible 3D details about the physical counterparts. Our study extends the concept of digital twin to replicate implicit motion and interaction features of dynamic agents, constructing a digital replica comprising of historical trajectories to lay the basis for robust and accurate motion forecasting.

3. Approach

3.1. Digital twin-based motion forecasting framework leveraging cloud-edge collaboration

In this paper, the digital twin technology is adopted to develop a motion forecasting model that enables risk monitoring and prediction on construction sites. The architecture of the proposed framework is illustrated in Fig. 1, comprising two main components: the physical world and the digital world. Besides, the edge-cloud framework is utilized to bridge the physical world and the digital world. It combines cloud computing and edge computing to enable efficient behavioral digital twin construction and accurate motion forecasting. In this framework, edge computing platforms provide real-time and low-latency processing for handling multi-view raw images captured in the physical world. The cloud computing platform receives the feature queues transmitted from the edge computing platforms, extracts motion information from the constructed digital description of surrounding context in BEV space, and perform motion forecasting exploiting its robust computing and storage capabilities.

In the physical world, multiple cameras are deployed to capture multi-view videos of the scene, which are processed by deep CNN-based backbones. In the digital world, a Transformer-based tracking model is devised to exploit the cross-attention mechanism, implementing a query-based approach to extract the motion features of workers in a unified global bird's-eye-view (BEV) space, and subsequently obtains historical trajectories to construct a fine-grained task-oriented behavioral digital replication of the physical world. Subsequently, a social LSTM-based model is designed to encode interactive features in both spatial and temporal dimensions, and predict the future trajectories of workers. The overall architecture of digital replica construction and motion forecasting is presented in Fig. 2.

3.2. Construction of the behavioral digital replica

The digital replica we construct in this study is a behavioral replication of workers' motions, effectively represented as historical trajectories. To accurately detect and track multiple workers based on features collected from multiple perspectives, we propose an instance-oriented query that encodes cross-view motion features by leveraging the cross-attention mechanism of Transformer, building upon the baseline model proposed by Zhang et al. [15]. The obtained query is then decoded into a 10-dimensional 3D bounding box, which captures the dynamic motion characteristics of the worker at the current frame in the BEV space. Moreover, the 3D bounding boxes of the same query across continuous frames within the observed time span naturally form a tracked trajectory, serving as the basis of the digital replication.

In order to represent workers' motion features in a global-refined 3D scene space for robust digital replica construction, we integrate the information from multi-view camera in the birds' eye view (BEV) space. First of all, we take the area covered by multiple cameras as the entire BEV space and divide it into grids of the same size. Each grid represents an area in the real world corresponding to an image region captured by multi-view cameras. Second, the center point of BEV space is set as the origin, and we perform joint calibration on all cameras to obtain the transformation matrix from pixel space to BEV space, including rotation matrix and translation matrix. Finally, we initialize some 3D reference points in BEV space to roughly represent the locations of target instances, which will be projected to multi-view image and the corresponding query aggregates temporal-spatial motion features via the cross-attention mechanism. Note that there are no explicit grids in the constructed BEV space. Instead, we just exploit the sparse reference point in BEV space and the corresponding query-based feature to represent the location and context-related motion feature of instance in construction scenarios.

Given multi-level feature maps extracted from multi-view images $F = \{F_1, F_2, F_3 \mid F_m = \{F_{m1}, F_{m2}, F_{m3}, F_{m4}\}, \mathrm{m} \in \{1,2,3\}\}$ where F_{mj} denotes the feature of level j extracted from images captured by camera m. Queries denoted as $q^t = \{q_1^t, q_2^t \ldots q_n^t\}$ are initialized by embedding

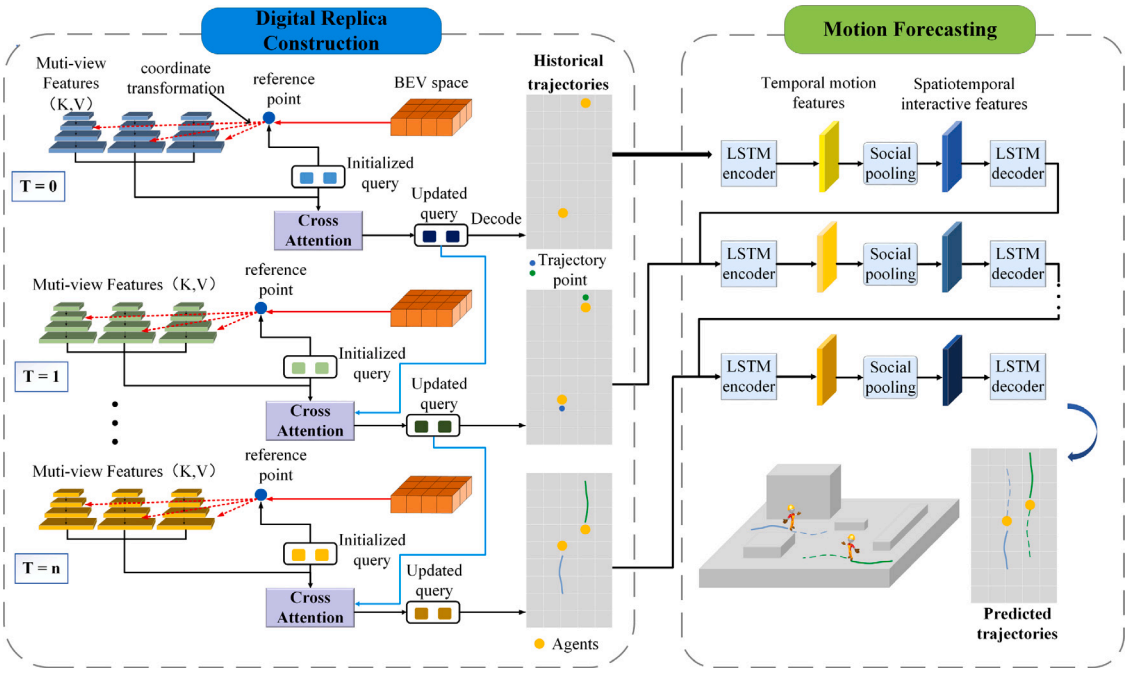


Fig. 2. The diagram of digital replica construction and motion forecasting.

layers, t means at time t, n represents the number of instances observed in the scene which varies as time goes by. 3D reference points c^t which depicts the position of agents in the BEV space are generated by queries. Then the sampling point set denoted as d^t, which contains various sampling points in the image pixel coordinate, are acquired via mapping the 3D positions from the BEV space to the image space. Exploiting the cross-attention mechanism, features of sampling points are aggregated to update that of the reference 3D point. This process can be expressed mathematically as follows.

$$c_i^t = \psi\left(q_i^t\right) \tag{1}$$

$$d_{i_j}^t = \pi_j\left(c_i^t\right), j = \{1,2,3\} \tag{2}$$

$$d^t = \left\{ d_1^t, d_2^t \ldots d_i^t \mid d_i^t = \left\{ d_{i_1}^t, d_{i_2}^t, d_{i_3}^t \right\} \right\} \tag{3}$$

$$f_{c_i}^t = \sum_{m=1}^{4} \sum_{j=1}^{3} \Phi\left(F_{mj}, d_{i_j}^t\right) \sigma\left(\omega_{mji}\right) \tag{4}$$

Where c_i^t represents the 3D reference point of query i, $\psi(.)$ denotes multi-layer perceptron, π_j is the mapping matrix from the BEV space coordinates to image pixel coordinates of camera j, d_i^t is the set of sampling points on feature maps of multi-view cameras for query i. Besides, $f_{c_i}^t$ denotes the aggregated features of the 3D reference point, $\Phi(.)$ stands for bilinear sampling, $\sigma(.)$ represents the sigmoid activation function, and ω_{mji} represents the weight to fuse features from sampling points. Based on the aggregated feature of 3D reference point, denoted as $f_{c_i}^t$, we can update both c_i^t and q_i^t.

$$q_i^t \Leftarrow q_i^t + \Psi\left(f_{c_i}^t + \Gamma\left(q_i^t\right)\right), c_i^t \Leftarrow c_i^t + \Psi\left(f_{c_i}^t\right) \tag{5}$$

Furthermore, the updated 3D reference point and query feature are decoded and a 10-dimensional 3D bounding box : $x_i^t \in R^3$ is the center of the 3D bounding box, $s_i = (w_i, l_i, h_i) \in R^3$ represents the size of the 3D bounding box, $v_i^t = \left(v_x^t, v_y^t\right) \in R^2$ is the velocity of the target in the pixel coordinate system, $o_i^t = \left(\sin\theta_i, \cos\theta_i\right)$ denotes the heading of the agent. The procedure can be expressed mathematically as below:

$$x_i^t = c_i^t + \Psi\left(q_i^t\right) \tag{6}$$

$$s_i^t, v_i^t, o_i^t = \Psi\left(q_i^t\right) \tag{7}$$

Where $\Psi(.)$ is the multi-layer perceptron mentioned above. Furthermore, we adopt delicately designed rules proposed in [13] for query life management, which adds new queries to denote new targets and eliminates old queries to represent vanishing targets in observed time span.

3.3. Digital twin-based motion forecasting

Based on the digital replica constructed, a motion forecasting model is developed to encode both spatial and temporal features shaping the motions of multiple agents in the scene. Due to the absence of geometric and topological structures in construction site scenes, graph-based scene representation is not suitable. In the previous section, we generated the historical trajectories of the agents in the BEV space. Therefore, we also convert the entire construction site scene into a rasterized BEV grid. Building upon this, inspired by the social LSTM proposed by Alahi et al. [16], we first utilize LSTM to extract temporal motion features from the target's historical trajectories. Subsequently, the social pooling mechanism is employed to model the spatial interactive relations among multiple agents, ultimately deriving a social tensor that characterizes the spatiotemporal interaction among them. This tensor is then decoded to generate predictions of future trajectories.

Specifically, the historical trajectories of workers are denoted as $X = \{X_1, X_2, \ldots, X_n \mid X_i = \{x_i^t \mid t \in \{t - T_{obs}, t - T_{obs} + 1, \ldots, t\}\}\}$ where i is the index of the worker and T_{obs} represents the length of recorded historical trajectories. Initially, the LSTM encoder embeds x_i^t as $r_i^t \in R^D$, where x_i^t is the motion information for agent i (the target agent) at time t, D denotes the dimension of hidden state. To represent interactions among multi-agents, we utilize a social tensor T_i^t to aggregate other agents' motion features in a predefined interactive region, which is a rectangle with the size of $\{M \times N\}$. The process can be expressed as below.

$$T_i^t \in R^{M \times N \times D}, T_{i_{(p,q)}^t}^t = \sum_{j \in I_{(p,q)}^t} h_j^{t-1} \in R^D \tag{8}$$

Here, $T_{i_{(p,q)}^t}^t$ is the social tensor of a single cell with coordinates (p, q) in the interactive region, h_j^{t-1} is the hidden state of agent j residing within

Table 1
Comparison with the baseline approaches on public dataset ETH/UCY using metrics ADE/FDE. The number of input and output frames is set to 12 and 8. Bold indicates the best.

ETH/UCY	ADE↓(m)/FDE↓(m)					
scenarios	Vanilla LSTM	Social-LSTM	Social-GAN	Trajectron++	Social-STGCNN	**ours**
ETH	1.09/2.41	1.09/2.35	0.87/1.62	0.71/1.66	**0.64/1.11**	0.77/1.88
Hotel	0.86/1.91	0.79/1.76	0.67/1.37	**0.22/0.46**	0.49/0.85	0.70/1.4
Univ	0.61/1.31	0.67/1.40	0.76/1.52	0.44/1.17	**0.44/0.79**	0.75/1.32
Zara1	0.41/0.88	0.47/1.00	0.35/0.68	0.30/0.79	0.34/0.53	**0.30/0.67**
Zara2	0.52/1.11	0.56/1.17	0.42/0.84	**0.23/0.59**	0.30/0.48	0.51/1.02

the cell at time $t-1$. Furthermore, interaction feature e_i^t is generated based on the influential tensor T_i^t.

$$e_i^t = Y\left(T_i^t, W_e\right) \tag{9}$$

Here, $Y(.)$ denotes the embedding function, W_e represents the parameters in the embedding function. Hidden state h_i^t is updated according to the formula below.

$$h_i^t = \mathrm{LSTM}\left(h_i^{t-1}, e_i^t, r_i^t, W_e\right) \tag{10}$$

Here, W_e denotes the LSTM weights. Once the final hidden state is obtained through multiple LSTM layers, the predicted trajectory of agent i denoted as $Y_i = \left\{y_i^t \mid t \in \left\{t+1, t+2, \ldots, t+T_{pre}\right\}\right\}$ is produced, where T_{pre} is the time span of the predicted trajectory. Predictions of all agents presented in the scene are obtained through the same process as described above.

4. Experiments and results

Our proposed digital twin-based motion forecasting model is trained utilizing two public dataset, ETH and UCY, which are widely used to train and test human motion forecasting approaches. We test the proposed model on a dataset collected from real-world construction sites in Wuhan, China. Both quantitative and qualitative analysis are conducted, and the results clearly demonstrate the effectiveness of our proposal in forecasting workers' motion robustly and accurately in various scenes on construction sites.

4.1. Dataset

The self-created test dataset WHUT-MF comprises 165 sequences, each containing 20 images captured by multiple cameras over a span of 5 s. The label for each sequence is represented as [pedestrian ID, x coordinate, y coordinate, z coordinate, predicted timestep], where the predicted timestep depends on both the prediction horizon and frequency. The accurate 3D positions (x, y, z) of multiple workers in the predicted time domain are measured using LiDAR and portable positioning sensors, with a sampling frequency matching that of the cameras.

For the purpose of validation, we specifically focus on challenging scenarios that might harbor hidden risks and pose difficulties for workers' motion forecasting. Examples include the vicinity of ascending or descending commodities suspended from tower cranes, the surrounding areas of slow-moving trucks, and the proximity to rotating crane counterweights.

4.2. Evaluation metrics

To evaluate the effectiveness of our proposed digital twin-based motion forecasting model, we utilize the following two commonly used metrics to measure the accuracy of motion forecasting: (1) Average displacement error (ADE): the average Euclidean distance between the real value of the target's position and its predicted value on all of the prediction steps. (2) Final displacement error (FDE): the Euclidean distance between the real value and predicted value of the target's position only on the final prediction step. Furthermore, we propose a new metrics named weighted displacement error (WDE), which is the weighted average of ADE and FDE. Given the nature of most accidents resulting from sudden latent risks, we intuitively prioritize FDE in assessing whether workers can timely avoid hazards. Specifically, we empirically assign a weight of 0.3 to ADE and 0.7 to FDE.

4.3. Implementation details

The proposed framework for edge-cloud collaboration is supported by the ROS operating system, ensuring well-organized and efficient communication between edges and the cloud. 3 cameras are securely mounted on a stationary platform in three perspectives: front, left, and right. To obtain 3D accurate positions of workers as ground truth labels, we employ the Alpha Prime sensor developed by Velodyne, which is installed on top of the stationary platform.

Edge computing platforms are equipped with cameras deployed on construction sites. The central computing platform is equipped with 4 high-performance Nvidia 3090 GPU. To facilitate real-time analysis, our system employs the MQTT (Message Queuing Telemetry Transport) protocol to transmit data streams generated by the edge computing platforms to the cloud.

For training of the motion forecasting model, the Adam optimizer with an initial learning rate of 0.001 is adopted. We use the same set of key hyperparameters as in [16], and set the training epoch to 30. The training platform is Intel(R) Xeon(R) Silver 4210 CPU @ 2.20 GHz, Nvidia GTX 3090 GPU with 32G RAM. It takes roughly four hours to train the model.

4.4. Quantitative evaluation

In Table 1, we show the performance of our proposed motion forecasting model on public datasets ETH and UCY, which are merged into ETH/UCY comprising five scenes (ETH, Hotel, Univ, Zara1, Zara2) . Results reveal that our approach achieves performance on par with SOTA (State of the Art) when predicting the movements of human-beings in open and complex scenes.

Furthermore, to demonstrate the ability of motion forecasting of our proposed model on specific construction sites, we have conducted experiments on the collected WHUT-MF. The results in Table 2 reveal that our approach can predict motions of workers accurately across various scenes in general. It is also suggested that the model performs the best in scene B. We attribute this discrepancy to the lack of inconspicuous motion characteristics among workers in Scenes A and C. In these particular scenes, workers tend to remain stationary or exhibit minimal movement for extended duration. Consequently, it becomes challenging to extract rich and expressive features from historical motion, resulting in a lower prediction accuracy compared to Scene B.

Furthermore, we conduct a series of experiments to examine the precision of the constructed digital replica. Since the digital replica is in fact composed of tracked historical trajectories of multiple objects, we adopt the metrics including average multi-object tracking accuracy (AMOTA) and average multi-object tracking precision (AMOTP). Additionally, we also aim to investigate the optimal time span of the historical trajectories to be utilized, which would not only enhance the accuracy of motion prediction but also lower the computational overhead associated with constructing the digital replica.

Table 2
Results of motion forecasting on the entire WHUT-MF dataset and 3 sets of hazardous scenes. Scene A, Scene B, and Scene C represent the vicinity of rising or falling commodities suspended from a tower crane, the surrounding areas of a slow-moving truck, and the neighborhood of rotating crane counterweights. These scenes are part of the WHUT-MF dataset.

	ADE↓ (m)	FDE↓ (m)	WDE↓ (m)
Entire WHUT-MF	0.528	0.841	0.747
Scene A	0.555	0.832	0.915
Scene B	**0.494**	**0.764**	**0.683**
Scene C	0.843	1.647	1.405

Table 3
Results of trajectory tracking and prediction with different time splits. α represents the time span of the historical data used to construct the digital replica; β is the prediction horizon, and their unit is second.

split($\alpha - \beta$)	AMOTA↑	AMOTP↓	WDE↓ (m)
split a(1-3)	0.241	1.593	0.839
split b(1.5-3)	0.249	1.587	0.819
split c(2-3)	**0.263**	**1.571**	**0.747**
split d(2.5-3)	0.256	1.579	0.793

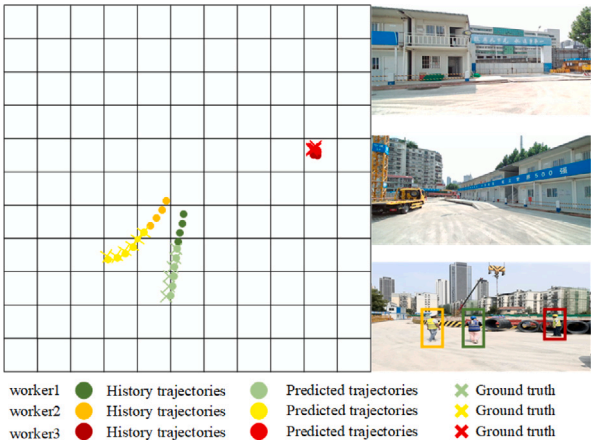


Fig. 3. Vicinity of lifting commodities.

We conducted experiments on the WHUT-MF dataset, utilizing different time splits, and the corresponding results are presented in Table 3. The analysis of AMOTA, AMOTP and WDE clearly demonstrates the influence of time span of historical trajectories on the accuracy of multi-object tracking, and consequently the precision of motion forecasting. It is also clearly suggested that the Weighted Displacement Error (WDE) varies in accordance with the accuracy of the constructed digital replica. From the experimental results, it is evident that longer duration of historical trajectories does not necessarily yield better performance. Our findings indicate that a 2-second duration of historical trajectories achieves the highest accuracy in motion tracking. Moreover, based on a 2-second historical trajectory, the motion prediction accuracy for the subsequent 3 s is also optimal.

4.5. Qualitative evaluation

In Figs. 3–6, we present visualized results of motion forecasting for four representative hazardous scenarios. The right column of each figure displays three raw images (from top to bottom) captured from the right, center, and left perspectives respectively. The left column showcases the historical trajectories tracked and future trajectories predicted for each worker presented in the corresponding scene. For better visualization, we project the 3D trajectories onto the BEV plane. Historical trajectories and predicted trajectories are represented by solid points, while ground truths are denoted by '×' symbols. Different colors are utilized to distinguish the motions of different workers.

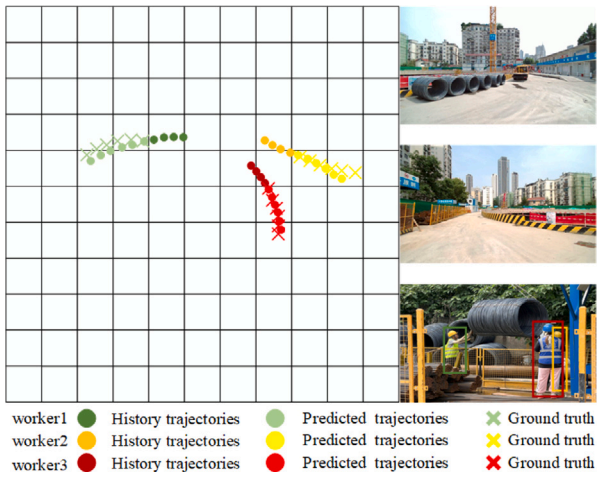


Fig. 4. Vicinity of delivering commodities.

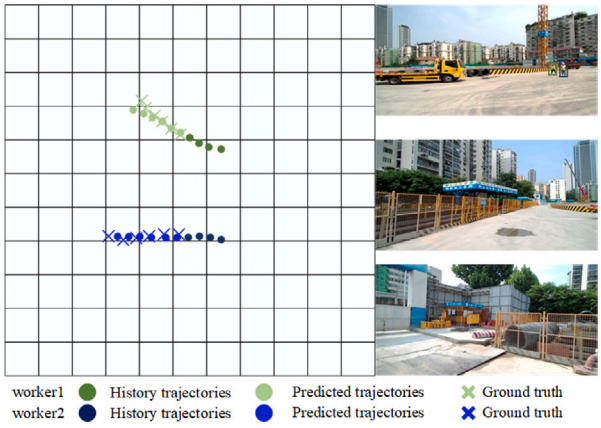


Fig. 5. Areas surrounding a slow-moving truck.

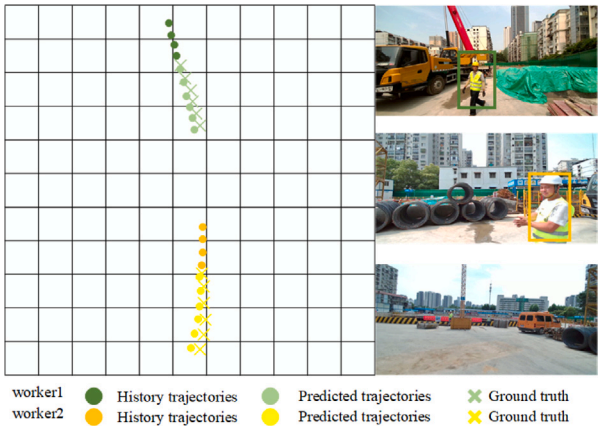


Fig. 6. Neighborhood of rotating crane counterweights.

Fig. 3 and Fig. 4 show workers gathering around a tower crane located in the foundation pit, where commodities are being lifted and delivered, respectively. The visualized predicted trajectories in Figs. 3–4 indicate that the workers' intentions of avoiding rising or falling heavy objects are effectively captured. Fig. 5 illustrates workers navigating through the field while avoiding an upcoming track. The results indicate that our approach successfully captures and represents both the workers' intention to avoid moving obstacles and the interactions

among them. Furthermore, the predicted trajectories aligns well with human behavior. Fig. 6 depicts a worker walking past a running crane, with special attention given to the rotating counterweights. Notably, our approach accurately forecasts the motions of workers by leveraging the digital replica and the interactive features among them, laying a solid foundation for effective and preemptive risk monitoring.

5. Conclusion

In this paper, a digital twin-based motion forecasting framework is developed and validated on real-world construction sites, laying a solid foundation for preemptive motion risk monitoring in service of intelligent safety management and provides a prototype for large-scale applications. In the physical world, cross-view images acquired by multiple cameras are processed with deep CNNs deployed on the edge-computing platform to obtain multi-scale and de-identification features about the scene. In the digital world hosted on the cloud, a fine-grained behavioral digital replication consisting of tracked trajectories of workers is constructed using a Transformer-based multi-object tracking model. Afterward, a social LSTM-based model is developed to capture the spatiotemporal interactive features, which are then decoded to generate future trajectories of workers. A series of experiments is designed and conducted to evaluate the effectiveness of the proposal. The results show that the proposed framework performs accurate and real-time motion prediction for multiple workers on construction sites.

Our work extends the concept of digital twin by incorporating human motion features into the digital replica and introduces a novel approach to monitor latent risks on construction sites. Future studies will focus on further exploiting digital twin technology to refine the construction of the digital replication with more detailed and nuanced information, such as worker's body movements, so as to integrate the recognition of workers' behaviors with motion forecasting to further enhance the accuracy of motion prediction. Additionally, multi-modal data instead of single-modal images could be used to further improve the model's robustness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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